



Crash Course in Maths for Cognitive Science

Session 5 | Probability & Statistics for Modelling

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What today is about

A refresher of the probability and statistics from your bachelor, brought back into working order and pushed one measured step further into the tools of inference.

- *Random variables*, the core *distributions*, *expectation* and *variance*.
- *Covariance*, joint distributions, and the *multivariate Gaussian*.
- The *Normal*, the *law of large numbers*, and the *central limit theorem*.
- *Estimation*, *maximum likelihood*, and *confidence intervals*.
- *Hypothesis testing* (z, t, two-sample), *Bayesian inference*, and a glimpse of modelling.

Definition

A **random variable** X assigns a number to each outcome of a random experiment. It is *discrete* if it takes countably many values (a die, a count of successes) and *continuous* if it takes values in a range (a height, a reaction time).

Intuition

The random variable is the “measurement”; the *distribution* is the rule that says how probable each value is. Everything else today is a property of that distribution.

How a distribution is described

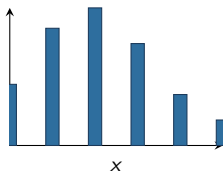
Useful fact

- **Discrete:** a *probability mass function* $p(x) = \mathbb{P}(X = x)$, with $\sum_x p(x) = 1$.
- **Continuous:** a *probability density* $f(x) \geq 0$ with $\int f(x) dx = 1$; probabilities are *areas*, $\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx$.
- **Either:** the *cumulative distribution* $F(x) = \mathbb{P}(X \leq x)$, increasing from 0 to 1.

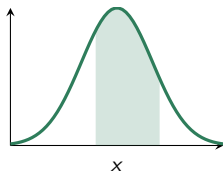
Common pitfall

For a continuous variable, $\mathbb{P}(X = x) = 0$ for any single point; only intervals carry probability. A density $f(x)$ can even exceed 1; it is an area, not a probability.

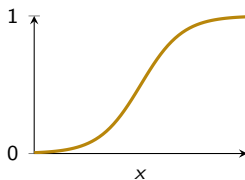
Three pictures: mass, density, cumulative



pmf (discrete)



pdf (area = probability)



cdf (climbs $0 \rightarrow 1$)

The cdf is the running total of the mass or area to the left, so it never decreases.

The distributions worth keeping at your fingertips

Distribution	Models	Mean / Variance
Bernoulli(p)	one yes/no trial	p / $p(1 - p)$
Binomial(n, p)	# successes in n trials	np / $np(1 - p)$
Poisson(λ)	# rare events in a window	λ / λ
Uniform(a, b)	“no value preferred”	$\frac{a+b}{2}$ / $\frac{(b-a)^2}{12}$
Normal(μ, σ^2)	sums of many small effects	μ / σ^2
Exponential(λ)	waiting time to an event	$\frac{1}{\lambda}$ / $\frac{1}{\lambda^2}$

The first three are discrete, the last three continuous. If you recognise which one a situation calls for, most problems become lookups.

The Binomial: try it

Example

A participant does 5 independent trials, each correct with probability $p = 0.4$. Let X be the number correct. What is $\mathbb{P}(X = 2)$, and what are the mean and variance of X ?

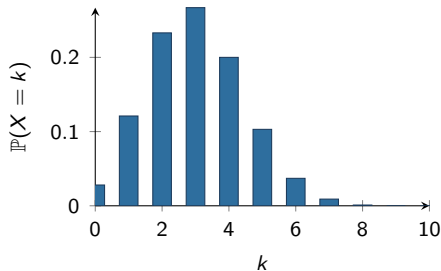
Try it yourself first — the solution is on the next slide.

The Binomial: solution

$X \sim \text{Binomial}(5, 0.4)$, so

$$\mathbb{P}(X = 2) = \binom{5}{2} (0.4)^2 (0.6)^3 \approx 0.35.$$

Mean $np = 2$, variance $np(1 - p) = 1.2$. The mass function is a bump peaking near the mean, as on the right (here $n = 10$, $p = 0.3$).



Binomial(10, 0.3) mass function.

From Binomial to Poisson, and waiting times

Key idea

When n is large and p small, with $\lambda = np$ moderate, the Binomial converges to the **Poisson**(λ), the law of counts of rare events in a fixed window (calls per minute, spikes per second).

Intuition

If events arrive as a Poisson process at rate λ , the *waiting time* between them is **Exponential** with mean $1/\lambda$. For $\lambda = 3$ per minute: $\mathbb{P}(\text{no event in a minute}) = e^{-3} \approx 0.05$, and the mean wait is 20 seconds. Counts are Poisson, gaps are Exponential, two sides of one coin.

Practice — try it

1. You flip a fair coin 10 times. Which distribution describes the number of heads, and what are its mean and variance?
2. A call centre receives on average 3 calls per minute. Which distribution fits the count in one minute, and which fits the gap between calls?
3. Is “reaction time in milliseconds” a discrete or a continuous variable?

Expectation: the long-run average

Definition

The **expectation** (mean) is the probability-weighted average value:

$$\mathbb{E}[X] = \sum_x x p(x) \quad (\text{discrete}), \quad \mathbb{E}[X] = \int x f(x) dx \quad (\text{continuous}).$$

Intuition

$\mathbb{E}[X]$ is where the distribution “balances”, the average you would approach if you repeated the experiment many times. It need not be a value X can actually take (the mean of a die is 3.5).

Variance: the spread

Definition

The **variance** is the average squared distance from the mean, and the **standard deviation** is its square root (back in the original units):

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mu^2, \quad \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

Intuition

Variance measures how widely values scatter around the mean. Two distributions can share a mean yet look completely different because their variances differ.

Useful fact

- **Linearity of expectation** (always, even for dependent variables):
 $\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$
- **Scaling/shift of variance:** $\text{Var}(aX + b) = a^2 \text{Var}(X)$; a shift does nothing, a scale squares.
- **Sums of *independent* variables:** $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$

Expectation and variance: try it

Example

Let X be the result of one roll of a fair six-sided die. Find $\mathbb{E}[X]$ and $\text{Var}(X)$.

Try it yourself first — the solution is on the next slide.

Expectation and variance: solution

Each face has probability $\frac{1}{6}$, so

$$\mathbb{E}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = 3.5.$$

For the variance, $\mathbb{E}[X^2] = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6}$, hence

$$\text{Var}(X) = \mathbb{E}[X^2] - \mu^2 = \frac{91}{6} - 3.5^2 = \frac{35}{12} \approx 2.92.$$

So $\text{SD}(X) \approx 1.71$, the typical distance of a roll from the average 3.5.

Definition

The **covariance** measures how two variables move together, and the **correlation** rescales it to $[-1, 1]$:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)], \quad \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}.$$

Intuition

Positive covariance: when one is above its mean, so is the other. Zero: no *linear* link. Correlation strips out the units and scale, so $+1$ is a perfect increasing line, -1 a perfect decreasing one.

Example

Four paired observations: $(1, 1)$, $(2, 3)$, $(3, 2)$, $(4, 4)$. Find the correlation of X and Y , and then the variance of $X + Y$. *Useful:* $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$.

Try it yourself first — the solution is on the next slide.

Covariance: solution

Both means are 2.5. Using $n - 1 = 3$: $\text{Var}(X) = \text{Var}(Y) = \frac{5}{3}$ and $\text{Cov}(X, Y) = \frac{4}{3}$, so

$$\text{Corr}(X, Y) = \frac{4/3}{5/3} = 0.8.$$

And $\text{Var}(X + Y) = \frac{5}{3} + \frac{5}{3} + 2 \cdot \frac{4}{3} = 6$.

Intuition

Because the two rise together (positive covariance), their sum varies *more* than if they were independent (which would give $\frac{5}{3} + \frac{5}{3} \approx 3.3$). Correlation of 0.8 is a strong but imperfect linear link.

Practice — try it

1. If $\text{Var}(X) = 4$, what is $\text{Var}(3X + 5)$?
2. X, Y independent with variances 2 and 3: what is $\text{Var}(X - Y)$?
3. If $\text{Corr}(X, Y) = 0$, must X and Y be independent?

Two variables at once

Definition

The **joint** distribution $p(x, y)$ gives the probability of pairs. From it, the **marginal** $p(x) = \sum_y p(x, y)$ describes one variable alone, and the **conditional** $p(y | x) = p(x, y)/p(x)$ describes one given the other.

Intuition

Marginal: ignore the other variable and sum it out. Conditional: fix the other variable and renormalise. The joint contains everything; marginals and conditionals are two ways of reading it.

Independence, and its weaker cousin

Key idea

X and Y are **independent** when $p(x, y) = p(x) p(y)$ for all values: knowing one tells you nothing about the other. Independence implies zero covariance.

Common pitfall

The converse fails in general: *uncorrelated does not mean independent*. Correlation only sees *linear* association, so a strong nonlinear link (for example $Y = X^2$) can have zero correlation. The one clean exception is the Gaussian, where uncorrelated *does* imply independent.

The covariance matrix

Definition

For a vector of variables, the **covariance matrix** Σ collects every variance and covariance:

$$\Sigma = \begin{pmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{pmatrix}.$$

It is *symmetric* and *positive-semidefinite* (no direction has negative variance).

Intuition

The diagonal holds the spreads, the off-diagonal the couplings. This is the exact object PCA diagonalises in Session 2: its eigenvectors are the principal axes of the data cloud.

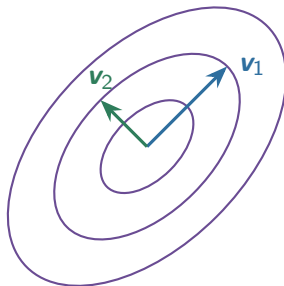
The multivariate Gaussian

Key idea

$\mathcal{N}(\mu, \Sigma)$ generalises the bell curve to several variables. It is fixed by a **mean vector** μ (the centre) and the **covariance matrix** Σ (the shape). Its equal-probability contours are *ellipses* whose axes are the eigenvectors of Σ and whose half-widths grow like $\sqrt{\lambda}$ of its eigenvalues.

That is the PCA picture from Session 2, now wearing a probabilistic hat: the covariance matrix is where statistics and linear algebra become one subject.

The covariance ellipse



Equal-probability contours of a 2-D Gaussian. The ellipse axes are the eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ of the covariance matrix; their lengths grow with $\sqrt{\lambda_1}, \sqrt{\lambda_2}$. Identical to the PCA picture.

Reading a covariance matrix: try it

Example

A 2-D Gaussian has covariance $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. In which directions do its contours stretch, and in what ratio? (*You found this matrix's eigenpairs in Session 2/4.*)

Try it yourself first — the solution is on the next slide.

Reading a covariance matrix: solution

The eigenpairs are $\lambda_1 = 3$ along $(1, 1)$ and $\lambda_2 = 1$ along $(1, -1)$. So the contour ellipses have their *long* axis along $(1, 1)$ and short axis along $(1, -1)$, with half-widths in ratio

$$\sqrt{\lambda_1} : \sqrt{\lambda_2} = \sqrt{3} : 1 \approx 1.73.$$

Intuition

Positive covariance tilts the cloud along $(1, 1)$: the two variables tend to rise and fall together. Eigenvector gives the tilt, eigenvalue the spread, reading the shape of data straight off Σ .

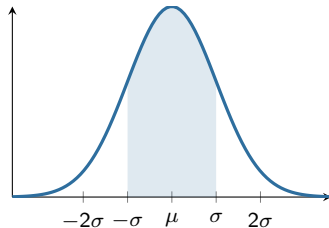
The Normal distribution

Definition

$X \sim \mathcal{N}(\mu, \sigma^2)$ has the bell-shaped density

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

μ sets the centre, σ the width. Its famous rule:
about 68% / 95% / 99.7% of the mass lies within
1 / 2 / 3 standard deviations of the mean.



The central 68% band ($\pm 1\sigma$).

Standardising: the z-score

Key idea

Subtract the mean and divide by the standard deviation to get a **standard normal** $Z \sim \mathcal{N}(0, 1)$:

$$Z = \frac{X - \mu}{\sigma}.$$

A z-score says how many standard deviations a value sits from the mean, so a single table of Z answers questions about *any* Normal.

The z-score: try it

Example

IQ scores are modelled as $\mathcal{N}(100, 15^2)$. What fraction of people score above 130?

Try it yourself first — the solution is on the next slide.

The z-score: solution

Standardise the threshold:

$$z = \frac{130 - 100}{15} = 2.$$

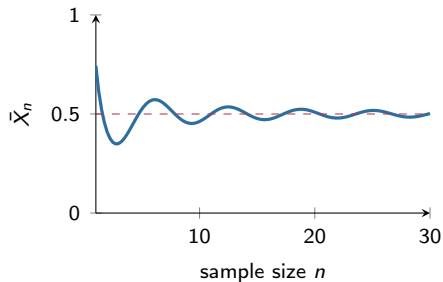
So we need $\mathbb{P}(Z > 2)$. From the 68–95–99.7 rule, 95% lies within $\pm 2\sigma$, leaving 5% in the two tails, or about 2.5% in each. The exact value is $\mathbb{P}(Z > 2) = 2.28\%$, roughly one person in forty.

The law of large numbers

Key idea

As the sample size grows, the **sample mean** $\bar{X}_n$ converges to the true mean μ . Averaging many observations cancels the noise.

This is why repeating and averaging works, and why a long-run relative frequency settles down to a probability.



A running average settling to the true mean.

The central limit theorem

Key idea

For large n , the sample mean is approximately **Normal**,

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right),$$

whatever the shape of the original distribution. Its standard deviation, the **standard error**, is $SE = \sigma/\sqrt{n}$.

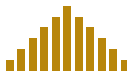
Intuition

The LLN says the average settles; the CLT says how fast and in what shape. This is the engine behind every confidence interval and test that follows.

The central limit theorem, in pictures



$n = 1$ (flat)



$n = 2$ (triangular)



large n (bell)

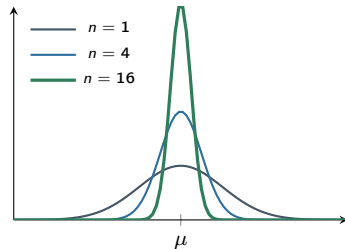
Averaging just a few draws already pulls a flat distribution toward the bell; the more you average, the closer it gets.

Why the standard error shrinks like $1/\sqrt{n}$

Intuition

Because $SE = \sigma/\sqrt{n}$, quadrupling the sample only halves the error; precision is expensive, but guaranteed. Note σ itself does not change with n , only the spread of the *mean* does.

The sampling distribution of the mean is (approximately) Normal with a known width, which is exactly what makes intervals and tests possible.



The mean's distribution narrows as n grows.

Practice — try it

1. A score is 1.5 standard deviations above the mean. What is its z-score, and is it in the top 10%?
2. Measurements have $\sigma = 15$. What is the standard error of the mean of $n = 25$ of them?
3. You average a very skewed variable over a large sample. What shape does the average have, approximately?

Definition

An **estimator** is a rule that turns a sample into a guess for an unknown quantity. The two everyday ones are the **sample mean** and **sample variance**:

$$\bar{x} = \frac{1}{n} \sum_i x_i, \quad s^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2.$$

Intuition

An estimate is itself random; a different sample gives a different number. So we care not just about the estimate but about how much it would wobble from sample to sample, its standard error.

Key idea

An estimator is **unbiased** if it is right *on average* over many samples. The sample mean is unbiased for μ . Dividing the squared deviations by n would *underestimate* the variance, because deviations are taken about the sample mean (which sits inside the data). Dividing by $n - 1$ corrects exactly for this.

The “ -1 ” is the one *degree of freedom* spent estimating the mean, and it comes back as the degrees of freedom of the t -distribution shortly.

Definition

Given a model $p(x | \theta)$ and independent data, the **likelihood** and its logarithm are

$$L(\theta) = \prod_i p(x_i | \theta), \quad \ell(\theta) = \sum_i \log p(x_i | \theta).$$

The **maximum-likelihood estimate** is the θ that maximises ℓ .

Intuition

Likelihood flips the usual question: fix the data we saw and ask which θ makes it most plausible. The log turns a product into a sum without moving the maximum.

Maximum likelihood: try it

Example

A participant succeeds on k of n independent trials, modelled as $\text{Bernoulli}(p)$. Find the maximum-likelihood estimate $\hat{p}$, and evaluate it for $k = 8$, $n = 10$.

Try it yourself first — the solution is on the next slide.

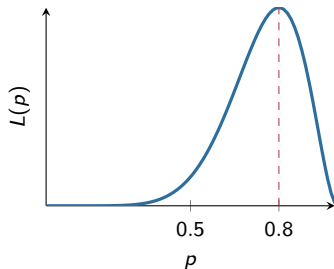
Maximum likelihood: solution

The log-likelihood is

$\ell(p) = k \ln p + (n - k) \ln(1 - p)$. Set $\ell'(p) = 0$:

$$\frac{k}{p} - \frac{n - k}{1 - p} = 0 \implies \hat{p} = \frac{k}{n} = 0.8.$$

Maximum likelihood recovers the obvious estimate, the observed success rate, but by a recipe that keeps working when the answer is not obvious.



$k = 8, n = 10$: peak at $\hat{p} = 0.8$.

Maximum likelihood recovers familiar estimates

Key idea

For Gaussian data, the MLE of the mean is the *sample mean* and the MLE of the variance is the *mean squared deviation*. And fitting a line by **least squares** is exactly maximum likelihood under Gaussian noise.

Example

For $\{2, 4, 4, 4, 5, 5, 7, 9\}$ the Gaussian MLEs are $\hat{\mu} = 5$ and $\hat{\sigma}^2 = \frac{32}{8} = 4$ ($\hat{\sigma} = 2$). The everyday summaries are maximum likelihood in disguise.

Confidence intervals (known σ)

Definition

A 95% **confidence interval** is a range built from the data so that, over repeated samples, 95% of such intervals contain the true parameter. For a mean with known σ ,

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} \quad (1.96 \text{ from the Normal}).$$

Common pitfall

The interval is random, not the parameter. “95% confidence” describes the *procedure* (95% of intervals catch the truth), not a probability that this one particular interval does.

A confidence interval: try it

Example

From $n = 25$ measurements with known $\sigma = 15$ you obtain $\bar{x} = 100$. Give a 95% confidence interval for the true mean μ .

Try it yourself first — the solution is on the next slide.

A confidence interval: solution

The standard error is $SE = \sigma/\sqrt{n} = 15/5 = 3$, so

$$100 \pm 1.96 \times 3 = 100 \pm 5.88 = [94.1, 105.9].$$

Intuition

Wider σ or smaller n widens the interval; more data shrinks it like $1/\sqrt{n}$. But this used a *known* σ , which we rarely have, that is what the t -distribution fixes next.

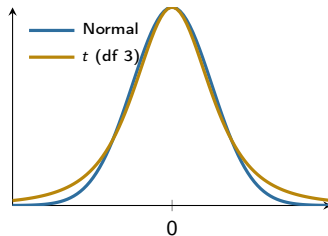
When σ is unknown: the t -distribution

Key idea

With σ unknown we estimate it by s , which adds uncertainty. The standardised mean then follows a t -**distribution** with $n - 1$ degrees of freedom, like the Normal but with *heavier tails*. The CI becomes

$$\bar{x} \pm t_{0.025, n-1} \frac{s}{\sqrt{n}}.$$

As n grows, t approaches the Normal (and $t_{0.025} \rightarrow 1.96$).



The t has heavier tails.

A t -interval: try it and check

Example

A sample of $n = 10$ reaction times has $\bar{x} = 52$ and sample SD $s = 6$. Give a 95% confidence interval for the mean. Use $t_{0.025, 9} = 2.262$.

The standard error is $s/\sqrt{n} = 6/\sqrt{10} \approx 1.90$, so

$$52 \pm 2.262 \times 1.90 = 52 \pm 4.29 = [47.7, 56.3].$$

Slightly wider than a z -interval would be (1.96 vs 2.262), the price of not knowing σ .

The logic of a test

Key idea

State a **null hypothesis** H_0 (“nothing is going on”) and an **alternative** H_1 . Assume H_0 , then ask: how surprising is the data we actually saw? If it is too surprising under H_0 , we reject H_0 .

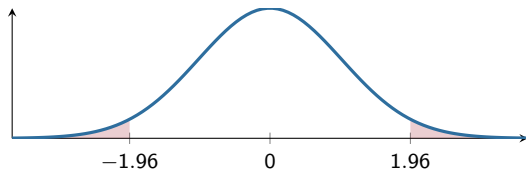
Intuition

A hypothesis test is a proof by contradiction with a probability attached. We never prove H_0 true; we only find the data compatible with it, or surprising enough to doubt it.

Test statistic, p -value, significance

Definition

A **test statistic** measures how far the data sit from what H_0 predicts, in standard-error units. The **p -value** is the probability, *if H_0 were true*, of a statistic at least this extreme. Reject H_0 when $p < \alpha$, the pre-chosen **significance level** (often 0.05).



Two-tailed test at $\alpha = 0.05$: reject if $|z| > 1.96$.

A one-sample z-test: try it

Example

A standard task has mean 100. A group of $n = 25$ people scores $\bar{x} = 106$; the population standard deviation is $\sigma = 15$. Test $H_0 : \mu = 100$ against $H_1 : \mu \neq 100$ at $\alpha = 0.05$.

Try it yourself first — the solution is on the next slide.

A one-sample z-test: solution

The standard error is $SE = 15/\sqrt{25} = 3$, so

$$z = \frac{\bar{x} - \mu_0}{SE} = \frac{106 - 100}{3} = 2.0.$$

Since $|z| = 2.0 > 1.96$, we reject H_0 at $\alpha = 0.05$ (two-tailed $p \approx 0.046$).

Intuition

Two standard errors above the null value is surprising enough, if nothing were going on, to doubt H_0 . But the group could still differ by a trivial amount: *statistical* significance is not *practical* importance.

The t -test: unknown σ

Example

A more realistic version: $n = 16$, $\bar{x} = 105$, and the *sample* SD is $s = 8$ (we do not know σ). Test $H_0 : \mu = 100$ at $\alpha = 0.05$.

The statistic uses s and the t -distribution with $n - 1 = 15$ df:

$$t = \frac{105 - 100}{8/\sqrt{16}} = \frac{5}{2} = 2.5, \quad t_{0.025, 15} = 2.131.$$

Since $2.5 > 2.131$, reject H_0 ($p \approx 0.025$). With a small sample and an estimated σ , the t -test is the honest choice.

Comparing two groups

Example

Group A ($n = 30$, mean 54) versus group B ($n = 30$, mean 48), each with SD 8. Is the difference significant?

The standard error of the difference is $\sqrt{8^2/30 + 8^2/30} \approx 2.07$, so

$$t = \frac{54 - 48}{2.07} \approx 2.9 \quad (p \approx 0.005).$$

Intuition

Comparing two means means testing whether their *difference* is far from zero, in units of the standard error of that difference. Variances add, so the two groups each contribute to the noise.

Two ways to be wrong

Useful fact

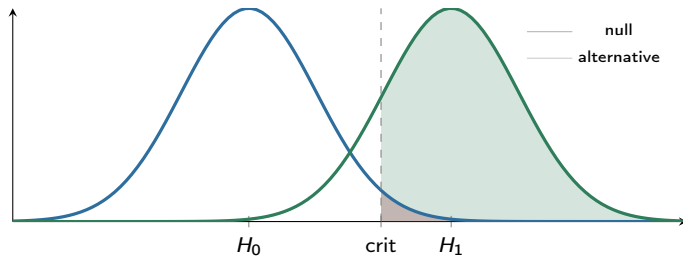
- **Type I (α):** reject H_0 when it is true, a false alarm.
- **Type II (β):** fail to reject H_0 when H_1 is true, a miss.
- **Power** = $1 - \beta$: the chance of catching a real effect; it grows with sample size and effect size.

	H_0 true	H_1 true
reject H_0	Type I	✓ power
keep H_0	✓	Type II

Common pitfall

A large p -value is *not* evidence that H_0 is true, only a failure to reject it.

Power, in a picture



Red: Type-I probability α (null past the critical value). Green: power (alternative past the critical value).
Moving H_1 further right, or shrinking the curves with more data, raises power.

Practice — try it

1. You double the sample size. By what factor does a confidence interval's width change?
2. When should you use a t -test rather than a z -test?
3. A test gives $p = 0.20$. Have you shown there is no effect? Name one way to raise power.

Conditional probability, and reversing it

Definition

The **conditional probability** of A given B is $\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$. Rearranged, $\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$.

Intuition

Often we know one direction, $\mathbb{P}(\text{positive test} \mid \text{disease})$, but want the other, $\mathbb{P}(\text{disease} \mid \text{positive test})$. Bayes' theorem flips the conditioning around.

Key idea

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A) \mathbb{P}(A)}{\mathbb{P}(B)}, \quad \mathbb{P}(B) = \mathbb{P}(B \mid A) \mathbb{P}(A) + \mathbb{P}(B \mid \bar{A}) \mathbb{P}(\bar{A}).$$

The denominator (the *total probability* of B) adds up all the ways B can happen. The numerator keeps the one path through A . Their ratio is the updated probability of A once we have learned B .

The base-rate example: try it

Example

A disease affects 1% of people. A test detects it in 99% of those who have it (sensitivity), but also gives a false positive in 5% of those who do not. You test positive. What is the probability you actually have the disease?

Try it yourself first — the solution is on the next slide.

The base-rate example: solution

Total probability of a positive test:

$$\mathbb{P}(+) = \underbrace{0.99 \cdot 0.01}_{\text{true } +} + \underbrace{0.05 \cdot 0.99}_{\text{false } +} = 0.0594.$$

Bayes' theorem then gives

$$\mathbb{P}(D | +) = \frac{0.99 \cdot 0.01}{0.0594} \approx 0.17.$$

Only about **17%**: the disease is rare, so most positives are false positives.

per 10,000	test +	test -
disease (100)	99	1
healthy (9900)	495	9405
total	594	9406

Of 594 positives, only 99 are truly ill:

$$99/594 \approx 17\%.$$

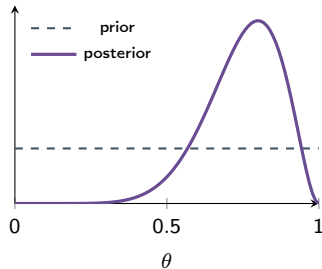
From events to a parameter

The same rule works when the unknown is a continuous *parameter* θ :

$$\underbrace{p(\theta \mid D)}_{\text{posterior}} \propto \underbrace{p(D \mid \theta)}_{\text{likelihood}} \times \underbrace{p(\theta)}_{\text{prior}}.$$

Intuition

Start with a prior belief over θ , reweight it by how well each value predicted the data, renormalise. The answer is a whole *distribution*: our belief after the data, uncertainty included.



Data sharpen a belief about θ .

Conjugacy: why Beta meets the Binomial

Put a $\text{Beta}(a, b)$ prior on a probability against binomial data of h hits and t misses. Multiply prior by likelihood:

$$\theta^{a-1}(1-\theta)^{b-1} \times \theta^h(1-\theta)^t = \theta^{(a+h)-1}(1-\theta)^{(b+t)-1} = \text{Beta}(a+h, b+t).$$

Key idea

The posterior is *another Beta*: just **add the hits and misses to the prior counts**. Read $a-1$, $b-1$ as “prior hits and misses”, so $\text{Beta}(1, 1)$ is a flat prior, “no information yet”.

Example

A detection probability θ has a flat prior $\text{Beta}(1, 1)$. You observe 8 detections in 10 trials. Find the posterior, its mean, and its peak (MAP).

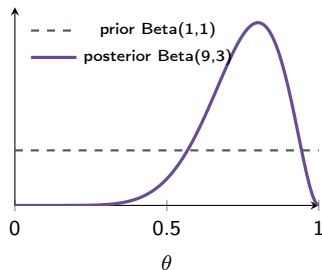
Try it yourself first — the solution is on the next slide.

Bayesian estimation: solution

Add 8 hits and 2 misses to the prior counts:
Beta(9, 3), with

$$\text{mean} = \frac{9}{12} = 0.75, \quad \text{MAP} = \frac{9-1}{12-2} = 0.80.$$

The peak sits at the MLE (0.8); the mean is pulled a little toward the prior centre. A 95% **credible interval** is about [0.48, 0.94]: we can say directly that θ lies there with 95% probability.



Flat prior to peaked posterior.

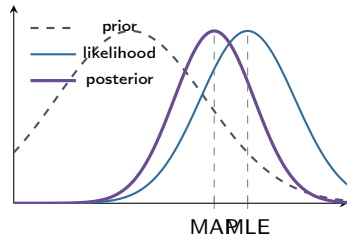
MLE, MAP, and the credible interval

Three summaries, one posterior:

- **MLE:** the peak of the *likelihood* (data alone).
- **MAP:** the peak of the *posterior* (likelihood $\times$ prior).
- **credible interval:** a range holding, say, 95% of the posterior.

Intuition

MAP is the MLE nudged by the prior (they coincide under a flat prior). Unlike a confidence interval, a credible interval is a genuine probability statement about θ .



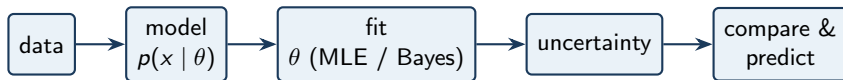
$$\text{Prior} \times \text{likelihood} = \text{posterior}.$$

Practice — try it

1. In the disease example, if the base rate rose to 10%, would $\mathbb{P}(D \mid +)$ go up or down?
2. Starting from $\text{Beta}(1, 1)$, what is the posterior after 3 hits in 3 trials, and its mean?
3. In one line, how does a credible interval differ from a confidence interval?

A glimpse ahead: modelling cognition

Everything so far *describes* data. A **cognitive model** goes one step further: it proposes *how* the data were generated, a mechanism with a few **parameters** we estimate from behaviour.



Intuition

The parameters *are* the psychology: a learning rate, a decision threshold, a perceptual bias. Distributions describe the data, likelihood fits the parameters, and the CLT, intervals, and Bayes quantify the uncertainty. You already own every piece.

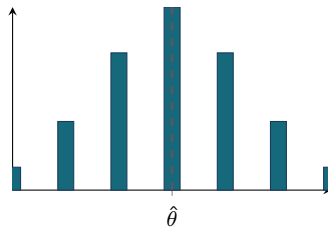
Uncertainty without formulas: the bootstrap

Key idea

When no neat standard-error formula exists, **resample the data with replacement** many times, recompute the statistic, and use the spread of those values as its standard error.

Intuition

The sample stands in for the population, so resampling from it mimics drawing fresh data. It works even for medians or correlations, where no textbook formula exists.



resampled statistic

Distribution of a statistic across resamples.

Session 5: what we covered

- *Random variables*, the core *distributions*, *expectation*, *variance*, and *covariance*.
- *Joint distributions* and the *multivariate Gaussian*, whose covariance matrix is the PCA/ellipse picture.
- The *Normal*, the *LLN* and *CLT*, with $SE = \sigma/\sqrt{n}$.
- *Estimation*, *maximum likelihood*, and *confidence intervals*; the *t-distribution* when σ is unknown.
- *Hypothesis testing* (z, t, two-sample), and *Bayesian inference* with priors, posteriors, MAP, and credible intervals.
- A glimpse of *modelling*, and the *bootstrap* for uncertainty.

These are the working tools you will reach for across data analysis, statistics, and the modelling courses ahead.